

Phrasebooks Can Teach Us ChatGPT: Decoding Prompt Crafting as Function Allocation

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Abstract

The discipline of Prompt Engineering is rapidly evolving, with varied and often conflicting approaches to crafting effective prompts for Large Language Models like ChatGPT. This divergence has led to a significant underutilization of AI technologies by individuals and organizations, largely due to a fundamental misconception about the nature of prompt writing. This paper advocates for a paradigm shift from a command-based to a language-centric approach, integrating concepts from metacognition and knowledge management. It introduces the novel metaphor of ‘Prompting Phrasebooks,’ akin to traveler phrasebooks, as a structured method to facilitate rich, interactive dialogues with AI systems. By drawing on a culturally ingrained analogy, the Prompting Phrasebook offers a practical framework for users with no or little expertise in prompt crafting to effectively engage with AI-driven chatbots. This concept is further operationalized through the creation of so-called ‘Prompt Pattern Languages,’ each representing a curated set of prompts designed to achieve a specific cognitive task. The paper presents the theoretical foundation of this approach and demonstrates its application through a Prompt Pattern Language for AI-assisted email composition, offering an unprecedented direction for teaching and mastering Prompt Engineering.

CCS Concepts

• **Human-centered computing**; • **Interaction design**; • **Interaction design theory, concepts and paradigms**;

Keywords

Large Language Models, Prompt Engineering, Prompt Pattern

ACM Reference Format:

Stefan Holtel. 2024. Phrasebooks Can Teach Us ChatGPT: Decoding Prompt Crafting as Function Allocation. In *29th European Conference on Pattern Languages of Programs, People, and Practices (EuroPLoP 2024)*, July 03–07, 2024, Irsee, Germany. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3698322.3698346>

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EuroPLoP 2024, July 03–07, 2024, Irsee, Germany

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ACM ISBN 979-8-4007-1683-6/24/07

<https://doi.org/10.1145/3698322.3698346>

1 The Advent of Natural Language Interfaces

The history of computer user interfaces has been marked by two paradigmatic shifts, each changing the way users interacted with technological progress (Table 1). With the advent of ChatGPT in late November 2022 user experience expert Jacob Nielsen predicted the arrival of the next interface paradigm [1]:

“ChatGPT and other AI systems are shaping up to launch the third user-interface paradigm in the history of computing — the first new interaction model in more than 60 years.”

The first paradigm, known as Batch Processing, emerged in the early 1940s. Users had to prepare complete workflows upfront, often using punched cards or tapes, with no opportunity for interaction during execution [2]. These workflows were processed in batches, typically during off-hours, and results were provided as printouts or punched cards. While this approach was methodical, it lacked flexibility and interactivity, making it suitable only for static data processing tasks.

The second paradigm, which arose in the 1960s, introduced Command Line Interfaces. Initially, teletypes printed each command entered by users, with responses also printed on paper. Those were replaced by terminals later, often referred to as ‘glass teletypes.’ This paradigm allowed for real-time interaction with the computer but surprisingly lacked a clear definition at its inception. Author and blogger Steven Johnson elaborates [3, p. 13]:

“This way of working did not even have a name at the time, but when, much later, an alternative became available, it was retroactively dubbed the Command Line Interface.”

Decades later, command lines became integral to personal computers, such as those using the Disk Operating Systems. This paradigm was defined by strictly turn-based interactions, where users input one command at a time. Unlike batch processing, it allowed for dynamic modification of goals after each round of input and output, offering more interactivity. However, there was a smooth flow of human-machine interaction. It is notable that the command-line paradigm persisted even with the advent of graphical user interfaces, like the Macintosh, while it dramatically altered the visual experience but retained the same fundamental principles [4] [3].

Entering the AI era, we now witness the next paradigm shift in computer interfaces. While it may appear to be a “rebirth of the command line interface” [5], this time it represents a significant departure. Now, AI interprets natural language prompts, shifting the focus from commands input to defining the computed outcome—a phenomenon Nielsen calls ‘Intent-Based Outcome Specification’ [6]. Instead of instructing the computer on how to perform a task, the user must clearly state the intended goal. Johnson would have

Table 1: Paradigms of Computer-Human-Interaction (enhanced from [1])

Batch-Processing	Command-Based-Interaction	Intention-Based Outcome Specification
Ca. 1940	From 1960s	Starting today
Specify the complete workflow upfront	Tell the computer how to do what you want	Tell the computer what you want
Deferred execution	Close to immediate feedback	Almost immediate feedback
Codified commands	Command prompts	Natural language prompts
Punched cards	Terminals and GUI	GUI and voice
Only machine-commands	Restricted to commands vocabulary	Dynamic goal modification

called it a new type of “the strange new zone between medium and message, we call it interface” [3, p. 45].

As we navigate this third paradigm shift in human-computer interaction, prompt engineering has emerged as a critical competency for leveraging its full potential. However, this radically new mode of interaction presents significant challenges for both individuals and organizations, who are often struggling to address it effectively. A McKinsey’s senior partner encapsulates this as follows [7]:

“Whenever you develop a technology, there will be a secondary effect somewhere in the system that will prevent you from fully capturing the value.”

The rapid and widespread deployment of AI chatbots is accompanied by several significant challenges. For example, a survey by [8] revealed that 48 percent of employees in Germany struggle to identify AI upskilling opportunities due to the overwhelming and confusing nature of market offerings. This highlights the need for a user-friendly approach to enable a broader adoption of AI chatbots.

Making use of generative AI chatbots is very different from using ‘traditional’ software [9]. Preliminary results show that for many people this becomes their first conscious encounter with AI systems. This means a significant shift in their hitherto understanding about the nature of the human-computer interface which leads to a surprising challenge how we must adapt and reinvent existing training concepts for such technologies [10]: Earlier computer interaction models relied heavily on predictable, linear, finite command exchanges. Chatbot interfaces instead introduce a highly unpredictable, open-ended commands interface that is optimized to simulate a human-to-human conversation, and even passed the Turing test [11].

This line of thought sheds light on the necessity of a deeper inquiry into human perception of AI systems, raising important questions how user’s existing cognitive models and prior experiences shape their engagement with an unprecedented technology.

2 In Search for Better Metaphors

Metaphors are not merely decorative elements of language but serve a fundamental role in shaping human thought and perception. Linguists Lakoff and Johnson interpret them as pervasive in everyday life, not just in language but even in thought and action [12]. They argue that metaphors influence how we understand abstract concepts by relating them to more concrete experiences. The realm of computers is not an exception but the rule in this regard. This discipline is saturated by metaphors. As user experience expert Robin Sease points out [13]:

“If metaphors exist as a necessary component of language – a tool to convey meaning and transfer information – then metaphors are by necessity a component of information science.”

Most computer users are not technically versed enough to fully grasp the inner workings of the computing systems they use daily: ‘files’ are verbatim ‘uploaded,’ ‘folders’ organize those ‘files,’ ‘folders’ hold ‘related data.’ But at the arrival of Large Language Models (LLM), those and many more metaphors become quickly outdated: chatbot data is not stored as ‘files’ but in ‘spaces,’ data is no longer organized in tagged ‘folders’ but in a concept called ‘distances between tokens representing meaning,’ folders therefore give way to ‘latent clusters’ [14]. Even the ultimate metaphor most users are stuck on becomes dubious: Computers are not ‘always right’ anymore. Instead, generative AI can ‘hallucinate’ in eerie ways—which is confusingly not a bug, but a feature [15] [16].

Most users remain anchored to outdated metaphors that have been in place for decades, now lacking cognitive models that adequately describe the functioning of these rapidly evolving technical systems. While new metaphors have emerged, such as innovation expert Ethan Mollick characterizing language models as ‘AI interns,’ these frameworks—though somewhat illustrative—mostly fail to provide users with practical guidance how to make use of them across diverse tasks [17]. This disconnect perpetuates the reliance on obsolete mental models, hindering users from fully capitalizing on the AI capabilities. There is an urgent need for fresh metaphors that empower users to much better engaging with LLMs. The idea of using ‘traveler phrasebooks’ could serve as a such a powerful metaphor.

2.1 A Short History of Traveler Phrasebooks

The history of travel phrasebooks can be traced back to ancient times when merchants and travelers created basic glossaries to communicate across language barriers. These early forms of phrasebooks facilitated trade and religious exchanges. Handwritten manuscripts provided essential phrases for pilgrims and traders, with Latin often serving as a lingua franca. The introduction of the printing press in the 15th century led to the publication of some of the first printed phrasebooks, such as William Caxton’s guide with English and French phrases for travelers [18]. The modern phrasebook began to take shape in the 19th century as middle-class leisure travel expanded across Europe. Authors like Karl Baedeker included sections of key phrases in their popular travel guides [19]. The 20th century saw a surge in phrasebook popularity with the rise of mass tourism, notably through practical phrasebooks,

which catered to travelers needing help with daily situations like ordering food or seeking directions [20]. Seminal works like the Lonely Planet phrasebooks, first introduced in the 1970s by John Wheeler, continue to be popular choices for budget-conscious and adventurous travelers seeking to navigate language barriers [21].

2.2 Phrasebooks as Pattern Languages

Architect Christopher Alexander invented the principles of Pattern Languages for a house building context [22], but they have meanwhile gained a longstanding tradition in software engineering [23]. It is unsurprising that software engineers would be among the first to adopt the idea of Pattern Language creation as a cognitive tool to improve prompt crafting skills. White et al. introduced a catalog of prompt patterns aimed at addressing common challenges in interactions with generative AI chatbots [24].

However, their work exhibits limitations. The theoretical framework of pattern languages is highly abstract for most LLM users. Additionally, the authors share a common background in software engineering, employing an overly technical approach to pattern elicitation. Furthermore, the paper presents a singular pattern language comprising sixteen patterns without suggesting methods for expanding those to other knowledge domains. All these constraints might hinder typical users to benefit from this groundbreaking work.

The contribution of this paper seeks to extend the foundational ideas presented, positioning itself as the next step towards rendering the well-established discipline of pattern languages into a versatile instrument for learning and teaching prompt crafting skills across diverse knowledge domains. The core concept is to compile so-called ‘Prompt Pattern Languages’ (PPL) as structured sets of patterns, each designed to encapsulate the most relevant prompts for achieving a specific objective by means of an AI chatbot. Envisioning a collection of several PPLs, users could open the ‘Prompting Phrasebook,’ and select a dedicated PPL from it for the communication or collaboration task they pursue by means of an AI chatbot. It is notable that this would be very different from a Prompt Library which is usually a weakly organized collection of prompts where prompts are loosely linked to each other and do not seek to precisely frame a pre-defined task. Mollick e.g. categorizes prompts just according to ‘Instructor Aids,’ ‘Student Exercises,’ and ‘Other Prompts’ [25].

Practically speaking, a prompting phrasebook is a curated collection of several pattern languages compiled for exploiting AI chatbot capabilities for versatile cognitive tasks. It provides pre-structured, reusable solutions to recurring communication and collaboration challenges encountered similar how travelers rely on phrasebooks to navigate challenging situations in foreign environments. Each PPL formalizes a bunch of ‘thinking pattern’ framing a specific cognitive task (e.g., conduct a brainstorming session, write a marketing report, outline workshop agenda) and offers reusable text phrases that can be readily employed in real-time interactions with LLMs. Chapter 4 will introduce a simple example of such a PPL.

3 Advance Organizers for Prompt Crafting

Building upon the discussion from the previous chapter, a critical question comes up: What should a Prompting Phrasebook entail,

how might prompts be structured and categorized? To address this, we must understand its purpose: It generally serves as a reminder for the user to express their own thinking processes. Such a phrasebook must therefore inherit a collection of cognitive tasks that reflect human thinking itself.

Theoretically speaking, this strongly aligns with development psychologist John H. Flavell’s important concept of ‘metacognition’—the process of thinking about one’s own thinking [26]. In the context of prompt engineering, the aim of prompts is to guide and streamline a human-machine communication in a clear, direct, and structured manner that enables the model to effectively respond to the input. Much like educational psychologist David Ausubel introduced ‘Advance Organizers’ to simplify complex information into manageable cognitive chunks [27], a PPL serves as cognitive instrument to help users learn and master advanced prompting techniques from powerful examples [28].

But before we can compile a PPL, there is a need to reframe the general understanding of Prompt Engineering [29]. In the following chapters, we will challenge three key assumptions: First, prompt engineering is not merely executing commands into a machine with a certain, expected outcome; it is rather a process of reducing huge uncertainty in the beginning through ongoing discourse. Second, prompt engineering is not reliant on a single, perfect prompt, but emerges from an iterative dialogue between the user and the LLM. Third, the final goal of prompt writing is not to automate sophisticated cognitive tasks, but to allocate mental workload strategically between the human and the machine.

3.1 Cone of Uncertainty

In the context of interactions with AI chatbots, the term ‘prompt’ refers to text input entered by the user into the command line prior to processing, while ‘completion’ denotes the AI’s response to that prompt [30]. The user possesses the agency to terminate the dialogue by choosing not to respond to the latest completion, thus characterizing the interaction as infinite. The interaction remains inherently open-ended, with the structure of the conversation shaped by the user’s willingness and ability to continue, typically executed through closed- or open-ended questions [31]. Beyond potential technical or performance flaws, an LLM is programmed to respond to any prompt, thereby ensuring that chatbots effectively maintain the final word.

Reducing uncertainty serves a significant role in human-to-human turn-taking communication [32], and it is reasonable to assume a similar dynamic exists in human-machine interactions. The user initiates the process with an ‘initial prompt’ that typically reflects a high degree of uncertainty regarding the topic under discussion (Figure 1, left).

The subsequent, increasingly nuanced chat interaction mitigates uncertainties, ultimately culminating in a ‘conclusive completion’ (Figure 1, right). Thus, the human-machine system traverses a ‘cone of uncertainty,’ where the final response determines whether the user experiences satisfaction (optimal outcome) or frustration due to unmet needs (suboptimal outcome). Engaging with a chatbot can be conceptualized as a gradual process of uncertainty reduction facilitated by intense turn-taking; each sequential interaction leads to narrow down uncertainty until the user perceives that his

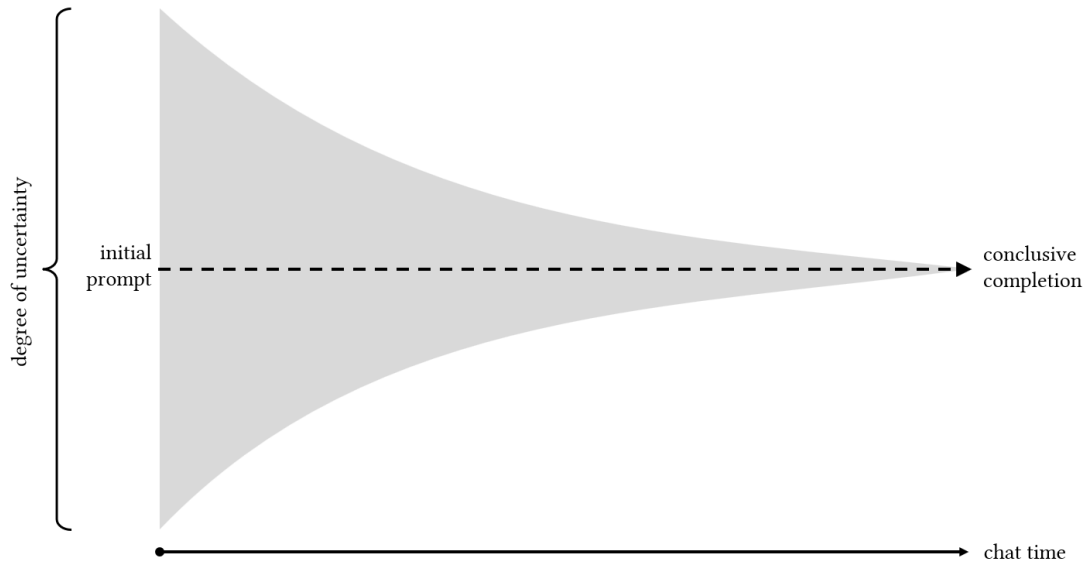


Figure 1: Cone of Uncertainty in Chatbot Dialogues

informational requirements have been fulfilled. What underlying intention drives such information-seeking behavior? According to communication expert Brenda Dervin the user’s objective can be framed as the mental process of ‘sensemaking’ [33]. Throughout a typical chat, the user navigates with curiosity and a divergent mindset, ultimately arriving at a sense of satisfaction by the end of the interaction. Turn-taking moderates the ebb and flow of data, information, and knowledge. Education psychologist Roy D. Pea would have posited that a chatbot functions as a cognitive system that reorganizes cognitive capabilities rather than merely amplifying them [34].

3.2 Turn-Taking

Misconceptions regarding the nature of prompting are prevalent at both individual and organizational levels. Most of the contemporary approaches to prompt crafting draw on concepts from the command line era, which are often outdated and inadequate for the current context. For instance:

- Software included user manuals for instructions and troubleshooting. Today, LLMs typically lack manuals, placing the onus on users to understand how to solicit assistance.
- Computers emphasized ‘commands’—short phrases modified with parameters for specific outcomes. In contrast, LLMs rely on ongoing dialogue in human natural language, refining user needs through each exchange.
- ‘Libraries’ spread to store useful commands for later use. This is less applicable to AI chatbots, which can respond to diverse natural language inputs in unpredictable ways.

A crucial aspect of understanding interactions with LLMs is the significance of turn-taking in dialogue. Drawing on the work

of Sacks et al., turn-taking in human conversation is governed by organizational principles that ensure orderly exchanges, with speakers taking turns predictably [32]. This process involves minimal units—such as sentences, clauses, or brief phrases—that facilitate conversational flow.

A comprehensive understanding of structured dialogue is essential for designing interactive technologies that emulate human conversational patterns with AI-driven chatbots. Such human-machine interactions face specific challenges. Tewari e.g. conducted research on breakdown situations in dialogues between humans and socially intelligent agents, proposing strategies and computational frameworks for managing those interactions as learning opportunities rather than errors [35].

Contrasting turn-taking between two humans to that between a human and a computer can be interpreted as an externalization of the user’s ‘internal dialogue’ [36] [37]. Inner dialogues, or self-talk, constitute the internal conversations individuals have with themselves, reflecting their thoughts, emotions, and reasoning processes. These dialogues are central to self-awareness, decision-making, and problem-solving, enabling individuals to process experiences, regulate emotions, and reflect on actions. Thus, ‘internal speech’ significantly shapes how individuals understand themselves and their environments—and interact with AI chatbots.

Conclusively, a successful interaction with an LLM is rarely the result of a single prompt; rather, it emerges from a dynamic back-and-forth between the user and the model. This iterative process is essential for shaping the conversation and ultimately achieving a desired outcome.

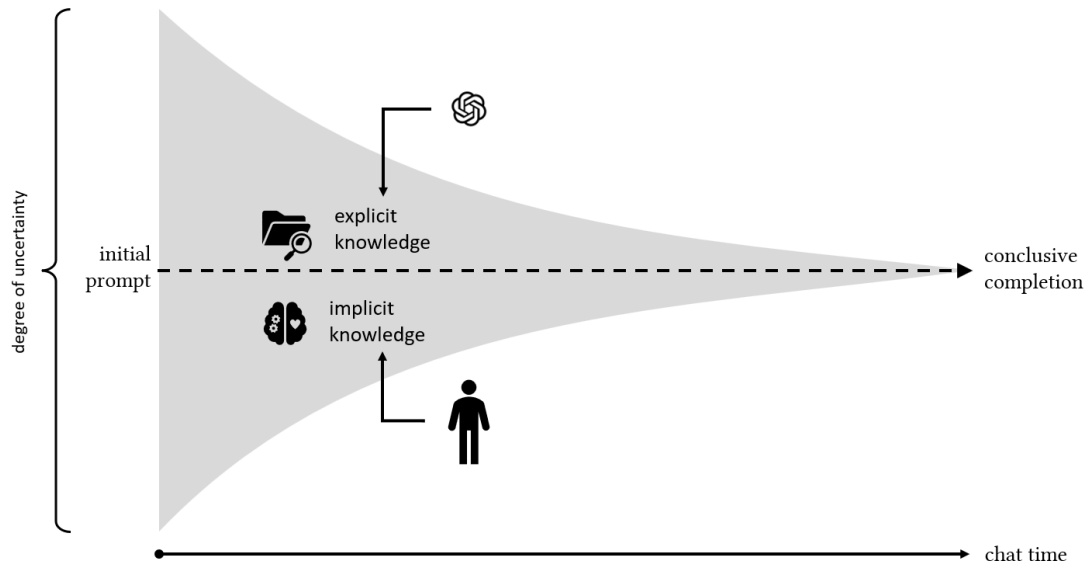


Figure 2: Knowledge in a Turn-Taking Dialogue

3.3 Allocation of Cognitive Workload

As education expert Roy Pea explored how cognitive functions are distributed between humans and technological tools, he introduced the term ‘function allocation’ [34]. He emphasized that technology does not merely aid humans but actively transforms cognitive tasks, which developed his ideas further into ‘distributed intelligence,’ highlighting that cognition is not confined to the individual but spread across people, tools, and environments, fundamentally altering the way humans think and solve problems [38]. LLMs can be considered as such tools in a distributed intelligence. But how are cognitive functions allocated between human and machine in this case? Here, knowledge theory comes to pass.

Each turn-taking between a user and a LLM implies the exchange of two possible types of knowledge. Knowledge can be categorized as being of implicit or explicit nature (Figure 2, upper and lower half of the cone).

- Explicit knowledge can be infused into a chat session through two possible pipes: it can be fed in during the chat session (e.g. paper, reports, meeting records, financial data) which is called ‘in-context-learning’, or it is already embedded in the huge training data set of an LLM.
- Implicit knowledge is of another nature. It emerges from the user while in the chat session. The user possesses knowledge from e.g. a long expert career or due to dedicated experiences accomplished otherwise (Figure 2, lower part of the cone). Polymath Michael Polanyi points out that this kind of knowledge cannot be codified for principled reasons. It would never make it into the course of a running chatbot dialogue without the user actively contributing.

For the purposes of this discussion, it is crucial to emphasize that explicit knowledge within the extensive LLM corpus, particularly in areas where there is minimal disagreement, often reflects long-established concepts. These ideas have been developed and validated over decades, if not centuries, and have proven effective in established practices. For instance, there is little doubt regarding the appropriate methodology for conducting a thorough root cause analysis [41] [42]. LLMs are capable to pour in such kind of knowledge especially for the purpose to guide users through a fruitful chat session while they will less suffer from hallucinations. We can describe this kind of knowledge as being ‘satisfied knowledge’ in terms of its usefulness and availability from a LLM corpus.

This analysis motivates the subject of the final chapter: How will a PPL precisely look like? How will it be compiled to ensure quality? Looking into the practice of creating one will make the approach more feasible. The following chapter provides a specific example of a PPL and unveils a few principles how it had been mined and compiled.

4 Example of a Prompt Pattern Language

After having outlined the theoretical approach to interpreting the discipline of prompt engineering as a metacognitive challenge, this chapter introduces a very simple example of a dedicated Prompt Pattern Language (PPL) that would reside in a Prompting Phrasebook comprising typical mental tasks of knowledge workers.

(Table 2) explores a recurrent daily cognitive challenge knowledge workers master many times a day: Writing an email. It consists of seven generic pattern (header row) and lists a bunch of possible prompts for each that are written in a way that a chatbot user can outsource parts of this mental task to an LLM while reducing his

Table 2: Prompts that Automate or Augment the Mental Effort of Writing Emails

Clarity	Structure	Tone	Accuracy	Actionability	Brevity	Appropriate Attachments
write email explaining [issue] benefits assist in refining the core message explain [issue] in simple language ask questions to ensure clarity	help structure content logically organize email outlining [issue] guide me to clear wording for non-tech readers break down project phases into sections and headings	draft an email with a formal tone for delays adjust the tone to be warm and professional make email friendly toned fine-tune tone for feedback	review numbers and dates for accuracy summarize email with accurate data check accuracy of {data, schedule} confirm [issue] details	outline {next steps, deadlines} in email encourage action without being too directive ask for feedback on the prototype clarify actions to be taken by the recipient	help edit email to be concise craft concise email with key milestones trim email while keeping details provide a brief status update on [issue]	ensure attachments are mentioned clearly reference the attached proposal in the email highlight attached [attachment] for meeting prompt recipient to review [attachment]
create an email explaining [issues]	format content for readability and flow	explain project changes to stakeholders	verify that data is accurate and updated	motivate team to meet project deadlines	revise email to be concise and direct	ensure attachments are clearly explained

cognitive load. The patterns have been derived by the process of first principle thinking to identify the key factors to accomplish this specific cognitive task.

The pattern names are organized based on a ‘flow of events’ of the cognitive task at hand, typically progressing from left to right. Each prompt can be classified as either automating a cognitive task (e.g., “highlight attached [attachment] for meeting,” “trim email while keeping details”) or augmenting it (e.g., “ask questions to ensure clarity,” “assist in refining core message”). While the number and names of the patterns should remain stable due to their backing in first principles, the specific number and wording of the prompts are preliminary and subject to change.

Square brackets, such as [issue] and [attachment], denote placeholders where specific content should be inserted which is relevant to the email’s context. For example, the placeholder [issue] suggests the insertion of a particular problem or topic that needs to be explained, while [attachment] indicate the need to reference a specific attached document. Curly braces, such as {next steps, deadlines} and {data, schedule}, encapsulate a set of related items or options from which one or more can be selected based on intended requirements. Those notations provide flexibility, allowing the content to be tailored to various scenarios appropriate for the audience.

The table functions as a ‘cheat sheet’ for novice LLM users, whereas more advanced users may view them as an inspiration, adapting and developing their additional prompts as needed, or initiate another PPL. The total number of prompts is flexible, as they may evolve based on practical experience and feedback, leading to either revisions, major overhauls, or even elimination. Furthermore, rapid technological advancements and new capabilities introduced in future LLM releases could necessitate changes to the prompts as well.

4.1 First Principles of Cognitive Workload

The critical step in developing a PPL lies in identifying the core cognitive task at hand. Without a deep comprehension of the actual mental challenge, prompt design risks becoming arbitrary. Thus, prompt engineering requires a reflective approach, wherein one must first analyze ‘thinking of thinking’ before iterating on outcomes to construct a meaningful PPL.

Several methods exist for conducting such an analysis. A prominent approach is the pursuit of ‘first principles’ [43] [44]. This problem-solving methodology emphasizes deconstructing complex issues into their most basic truths, allowing for the development of effective prompts from the ground up. A PPL serves as a reflective tool, encapsulating best practices in critical thinking [45]. Rather than depending on preconceived notions, first principles thinking encourages the creation of perspectives grounded in fundamental facts—an intellectually demanding task. Physicist Richard Feynman was well known for his favor of first principles as he said [46]:

‘I learned very early the difference between knowing the name of something and knowing something. One’s a definition and the other is understanding.’

In recent years, this approach has been notably popularized by Elon Musk, who has applied it to drive innovation in fields such as aerospace and electric vehicles. For instance, in the development of SpaceX, Musk challenged traditional assumptions about the prohibitive costs of spaceflight by deconstructing the problem to its fundamental components, specifically the raw materials used in rocket production, ultimately achieving significant cost reductions [47]. A preliminary first principles set of email writing might look like this:

- Clarity: Craft a subject line and files and ensuring they are correctly labeled
- Structure: Organize the email with a salutation, introduction, body, conclusion, and closing

- **Tone:** Adopt a professional tone, using respectful language, and tailoring the tone to the recipient
- **Accuracy:** Ensure the email is free from spelling and grammatical errors, with all information accurately presented
- **Actionability:** Provide a clear call to action or next steps, along with deadlines
- **Brevity:** Maintain brevity while ensuring the message is adequately conveyed
- **Appropriate Attachments:** Attach only necessary applied to seemingly straightforward tasks

The more effort we dedicate to analyzing a given task through the lens of first principles, the greater the quality of insights. Only a thorough analysis fosters deeper understanding of the fundamental elements of a complex mental task, allowing assumptions, biases, and superficial layers to be stripped away. As the depth of analysis increases, the clarity and precision of the principles identified improve, resulting in a valuable PPL that is grounded in solid, reliable foundations rather than conventional wisdom.

4.2 Automation vs. Augmentation Prompts

An often-cited approach to AI integration at workplace is that systems can either automate or augment tasks [48] [49]. Automation typically refers to AI taking over repetitive or routine tasks, while augmentation involves AI enhancing human capabilities in more complex tasks, such as decision-making, creative processes, or problem-solving. This dual role of AI—automating routine operations and augmenting human capacity—is a key element of AI implementation and can clearly be adapted to crafting valuable prompts for a PPL. A prompt can either automate the accomplishment of cognitive task or augment the user to accomplish a goal by a turn-taking, shared process.

Economists John McAfee and Eric Brynjolfsson emphasize the transformative potential of AI in reshaping both the nature of work and the economy, highlighting the dual dynamic between automation and augmentation. They argue that while automation can lead to increased productivity by eliminating certain routine jobs, only augmentation allows humans to leverage AI as a tool to enhance their own capabilities [50]. They argue that machines are particularly well-suited for tasks that are rule-based and predictable, whereas humans excel in areas requiring creativity, complex problem-solving, and adaptability. The integration of AI into the workplace, therefore, must recognize this complementarity, enabling AI systems to automate repetitive tasks while augmenting human ingenuity in higher-order activities. Such a balance, they suggest, is essential for navigating the digital transformation in a way that fosters both innovation and human growth. Although Brynjolfsson sees more potential in augmentation by stating [51]:

“So in most cases, it’s not a replacement but more of an amplification.”

In automation, the system performs a specific task autonomously, requiring no further input or intervention during its execution (which closely mimics our current understanding of computer commands) (Table 3, left). This may include tasks such as organizing data or executing pre-defined actions without manual oversight. They require no or minimal user interaction, relying mostly on declarative or procedural knowledge [52].

Augmentation involves the system assisting the user in decision-making or task execution by providing context, answers to additional queries, or suggestions on how to approach and prioritize problems (Table 3, right). Augmented interactions are especially poised to improve a user’s ability to analyze and synthesize information. Augmentation prompts are designed to assist or enhance the user’s cognitive processes, helping with understanding, decision-making, or creativity. These prompts involve active collaboration between the user and the model, focusing on providing insights, guidance, or explanations that improve the user’s own capabilities rather than delivering a final product. In this case knowledge management expert David Gurteen interprets AI chatbots as ‘critical thinking partners’ [53].

This categorization of automation and augmentation can be effectively applied to the creation of prompt patterns in LLMs. Prompts, which guide the system’s behavior through structured input, can either be designed to automate certain tasks or to augment them. For example, a prompt designed to automate a task might instruct the LLM to ‘format content for readability and flow’ completing the action without further user interaction. In this case, the user is freed from manual effort as the LLM efficiently organizes the data as requested. Automation here leads to increased productivity and consistency, especially in repetitive tasks.

In contrast, prompts that augment a task would make possible more complex cognitive activities. For instance, a prompt might ask the LLM to ‘guide me to clearer wording for non-tech readers’. This type of augmentation helps the user tackle more abstract problems, such as understanding context or improving communication strategies. By providing insights or options, the LLM augments the user’s capacity to make informed decisions, thereby enhancing the quality of output without removing the user’s agency in completing the task.

In linking this discussion to philosopher John Searle’s Speech Act Theory, we can consider the role of directives in shaping how prompts are crafted for AI systems, particularly with a focus of automation and augmentation. Searle identified directives as speech acts in which the speaker attempts to get the hearer to do something, such as requesting, commanding, or advising [54]. When we design automation prompts, they often take the form of directives that give clear, actionable instructions to the AI, leaving little room for ambiguity or interpretation. These prompts are akin to issuing commands, where the system is expected to execute a task without further engagement or input. On the other hand, in augmentation prompts, the directive nature becomes more nuanced. Instead of merely requesting a task completion, the user’s prompts aim to guide the system in supporting the user’s cognitive efforts, much like an advisor or collaborator. The LLM might respond with suggestions or clarifications that require further dialogue, turning the interaction into a ‘shared process of meaning-making’ rather than a simple command-following exercise. This distinction aligns with Searle’s view of language as action, where the nature of the directive varies depending on whether the goal is to automate (a straightforward request) or augment (a collaborative endeavor).

Table 3: Automation vs. Augmentation Prompts

Automation Prompts	Augmentation Prompts
involves executing predefined rules or algorithms	involves guiding, assisting, or enhancing user’s abilities
produces a concrete, result or product	produces insight, suggestions, or knowledge enhancement
minimal user engagement beyond providing input	active user engagement, involving clarification or collaboration
applies factual or procedural knowledge	applies reasoning, conceptual understanding, or creativity
focused on efficiency and task completion	focused on improving understanding or decision-making processes
requires little to no interpretation	requires judgment, interpretation, and contextual understanding

5 Conclusion and Outlook

This paper has explored the emerging challenges within prompt engineering, advocating for a transition from command language-driven to human language-centered approaches in prompt design. By integrating metacognitive frameworks and pattern languages, the concept of a ‘Prompting Phrasebook’ was introduced as a practical, structured guide for users. Much like travel phrasebooks assist in navigating unfamiliar languages and cultures, the Prompting Phrasebook offers a glossary for constructing task-specific, highly effective prompts, leveraging Pattern Language principles to enhance interactions with generative AI systems.

The findings emphasize the importance of deepening our understanding of prompt engineering to better manage cognitive load and to facilitate clearer communication through structured turn-taking in human-AI dialogues. A metacognitive approach addresses the often-misaligned expectations between users and AI capabilities, fostering more transparent and effective interactions.

This work aspires to serve as a guide for developing systematic methods to create and curate prompt collections tailored to a wide range of tasks and expertise levels. While AI chatbots might threaten critical thinking [55], this paper argues that they can be made more accessible, efficient, and impactful. The goal is to democratize the use of AI across diverse domains, enabling practitioners from various fields to benefit from AI’s capabilities while maintaining intellectual rigor. Future research should focus on refining prompt engineering methodologies and exploring how these systems can continue to evolve in support of critical and creative thinking across disciplines.

Acknowledgments

This research has been profoundly shaped by numerous inspirational sources and supportive communities. It is the culmination of a two-year intellectual exploration into the rapidly evolving field of prompt engineering, initiated at the close of 2022 with the public release of ChatGPT. This technology, far from being a transient phenomenon, signifies a disruptive breakthrough that will continue to influence our lives for years to come. I extend my deepest gratitude to the vibrant community of early adopters and scholars who have ventured into the uncharted territory of large language models. Their insights and enthusiasm have greatly enriched my understanding of the broader implications of prompt engineering and prompt crafting as an emerging discipline. Special thanks are due to OpenAI for their pioneering technologies and visionary approach, which not only inspired my recent book but also laid

the groundwork for further exploration into human-centered AI—a research frontier that underpins this paper and will guide future work.

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